



AMITY UNIVERSITY PUNJAB, MOHALI

B.SC. MATHEMATICS (5TH SEMESTER)
SET THEORY AND METRIC SPACES (MAT304)

Unit I: Theory of Sets

Practice Problems

Dr. Lokesh Singh
`lsingh@pb.amity.edu`

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Problems

Very Easy Problems

Problem 1. Use the Schröder–Bernstein theorem to prove that $|(0, 1)| = |(0, 1]|$. Explicitly define injections in both directions.

Problem 2. Construct an explicit bijection between the set of odd positive integers and $\mathbb{N}$.

Problem 3. Show that the set of all rational numbers in the interval $(0, 1)$ is countable.

Problem 4. Let $A = \{a, b, c, d\}$. Write down $P(A)$ and compute $|P(A)|$.

Problem 5. Give an example of a situation where the Axiom of Choice is required to define a choice function. Explain why the choice cannot be made constructively.

Medium Problems

Problem 6. Prove that $\mathbb{Q} \times \mathbb{Q} \times \mathbb{Q}$ is countable.

Problem 7. Prove that the set of all finite sequences of natural numbers (i.e., $\mathbb{N}^{<\omega} = \bigcup_{n=1}^{\infty} \mathbb{N}^n$) is countable.

Problem 8. Let A and B be countable sets. Prove that $A \times B$ is countable.

Problem 9. Show that the set of irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ is uncountable. Use the fact that $\mathbb{Q}$ is countable and $\mathbb{R}$ is uncountable.

Problem 10. Construct injections both ways between $\mathbb{R}$ and $\mathbb{R} \times \mathbb{R}$. Use the Schröder–Bernstein theorem to conclude that the line and the plane have the same cardinality.

Problem 11. Compute the following cardinalities:

- (a) $\mathfrak{c}^3$
- (b) $\mathfrak{c}^{\aleph_0}$
- (c) $(2^{\mathfrak{c}})^{\aleph_0}$

Problem 12. Let A be an infinite set. Prove that $|A| < |P(A)|$ using Cantor's theorem. Explain why this implies there is no largest cardinal number.

Problem 13. Let $f : A \rightarrow B$ be a function. Suppose there exists $g : B \rightarrow A$ such that $g \circ f = i_A$. Prove that f is injective.

Problem 14. Explain how the Axiom of Separation in ZFC prevents Russell's paradox. Give a concrete example of a collection that cannot be a set.

Problem 15. Use Zorn's lemma to prove that every partially ordered set has a maximal chain. (You may assume Zorn's lemma as an axiom.)

Hard Problems

Problem 16. Prove that the set of all permutations of $\mathbb{N}$ (bijections from $\mathbb{N}$ to itself) has cardinality $\mathfrak{c}$. Hint: Construct an injection from $\{0, 1\}^{\mathbb{N}}$ into the set of permutations by swapping pairs.

Problem 17. Show that the set of all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ has cardinality $\mathfrak{c}$. Use the fact that a continuous function is determined by its values on $\mathbb{Q}$.

Problem 18. Prove that the Axiom of Choice implies Zorn's Lemma. (Give the transfinite induction proof using the choice function to construct a strictly increasing chain indexed by ordinals.)

Problem 19. Let κ be an infinite cardinal. Prove that $\kappa \cdot \kappa = \kappa$. Use Zorn's lemma and the maximality argument on the set of partial bijections $E \rightarrow E \times E$.

Problem 20. Prove that $\mathfrak{c}^{\aleph_0} = \mathfrak{c}$ and $\mathfrak{c}^{\mathfrak{c}} = 2^{\mathfrak{c}}$. Use these results to compare the cardinalities of $\mathbb{R}^{\mathbb{N}}$ and $\mathbb{R}^{\mathbb{R}}$. Conclude that there are strictly more functions from $\mathbb{R}$ to $\mathbb{R}$ than from $\mathbb{N}$ to $\mathbb{R}$.